

● MEDIUM

● ALGEBRA

● ANSWER KEY

Linear equations in one variable

01

10 answers · Rationale-rich · Teach from doc

Q1**C****C.** 30

Choice C is correct. Subtracting 16 from both sides of the given equation yields $30 = 2(x - 8)$. Therefore, the value of $2(x - 8)$ is 30.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**A****A.** 2

Choice A is correct. This equation has no solution when there is no value of x that produces a true statement. Solving the given equation for x by dividing both sides by $(b - 2)$ gives

$x = \frac{8}{(b - 2)}$. When $(b - 2) = 0$, the right-hand side of this equation will be undefined, and the equation will have no solution. Therefore, when $b = 2$, there is no value of x that satisfies the given equation.

Choices B, C, and D are incorrect. Substituting 4, 6, and 10 for b in the given equation yields exactly one solution, rather than no solution, for x . For example, substituting 4 for b in the given equation yields $(4 - 2)x = 8$, or $2x = 8$. Dividing both sides of $2x = 8$ by 2 yields $x = 4$. Similarly, if $b = 6$ or $b = 10$, $x = 2$ and $x = 1$, respectively.

Q3**2**

The correct answer is 2. An equation with one variable, x , has infinitely many solutions only when both sides of the equation are equal for any defined value of x . It's given that $2x + 16 = ax + 8$, where a is a constant. This equation can be rewritten as $2x + 8 = ax + 8$. If this equation has infinitely many solutions, then both sides of this equation are equal for any defined value of x . Both sides of this equation are equal for any defined value of x when $2 = a$. Therefore, if the equation has infinitely many solutions, the value of a is 2.

Alternate approach: If the given equation, $2x + 16 = ax + 8$, has infinitely many solutions, then both sides of this equation are equal for any value of x . If $x = 0$, then substituting 0 for x in $2x + 16 = ax + 8$ yields $2(0) + 16 = a(0) + 8$, or $16 = 8a$. Dividing both sides of this equation by 8 yields $2 = a$.

Q4**D****D. 44**

Choice D is correct. It's given that the bowl starts with 20 ounces of water and has 9 ounces of water remaining after a period of time has passed. The amount of water the bowl has lost during the time period can be found by subtracting the remaining amount of water from the amount of water the bowl starts with, which yields $20 - 9$ ounces, or 11 ounces. This means the bowl loses 11 ounces of water during that period of time. It's given that the amount of water decreases by 1 ounce every 4 days. Letting t represent the number of days the bowl has been uncovered, it follows that $\frac{1}{4} = \frac{11}{t}$. Multiplying both sides of this equation by $4t$ yields $t = 44$. Therefore, the bowl has been uncovered for 44 days.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of t for the equation $\frac{1}{4} = \frac{9}{t}$, not $\frac{1}{4} = \frac{11}{t}$.

Q5

—

Answer not provided in source data — see rationale below.

Choice B is correct. Since p represents Megan's regular pay per hour, $1.5p$ represents the pay per hour in excess of 8 hours. Since Megan worked for 10 hours, she must have been paid p dollars per hour for 8 of the hours plus $1.5p$ dollars per hour for the remaining 2 hours.

Therefore, since Megan earned \$137.50 for the 10 hours, the situation can be represented by the equation $137.5 = 8p + 2(1.5)p$. Distributing the 2 in the equation gives $137.5 = 8p + 3p$, and combining like terms gives $137.5 = 11p$. Dividing both sides by 11 gives $p = 12.5$.

Therefore, Megan's regular wage is \$12.50.

Choices A and C are incorrect and may be the result of calculation errors. Choice D is incorrect and may result from finding the average hourly wage that Megan earned for the 10 hours of work.

Q6**C****C. 8**

Choice C is correct. If an equation has infinitely many solutions, then the two sides of the equation must be equivalent. Multiplying each side of the given equation by 2 yields $8x + 24 = ax + 3$. Since 8 is a common factor of both terms on the left-hand side of this equation, the equation can be rewritten as $8x + 3 = ax + 3$. The two sides of this equation are equivalent when $a = 8$. Therefore, if the given equation has infinitely many solutions, the value of a is 8.

Alternate approach: If the given equation, $4x + 12 = \frac{ax+3}{2}$, has infinitely many solutions, then both sides of this equation are equal for any value of x . If $x = 0$, then substituting 0 for x in the given equation yields $4(0) + 12 = \frac{a(0)+3}{2}$, or $12 = \frac{3}{2}a$. Dividing both sides of this equation by $\frac{3}{2}$ yields $8 = a$.

Choice A is incorrect. If the value of a is 0, the given equation is equivalent to $4x + 12 = 0$, which has one solution, not infinitely many solutions.

Choice B is incorrect. If the value of a is 3, the given equation is equivalent to $4x + 12 = \frac{3x+3}{2}$, or $4x + 12 = \frac{3}{2}x + \frac{3}{2}$, which has one solution, not infinitely many solutions.

Choice D is incorrect. If the value of a is 12, the given equation is equivalent to $4x + 12 = \frac{12x+3}{2}$, or $4x + 12 = 6x + 1.5$, which has one solution, not infinitely many solutions.

Q7**A****A. 2**

Choice A is correct. Adding the like terms on the left-hand side of the given equation yields $5(x-5) = 10$. Dividing both sides of this equation by 5 yields $x-5 = 2$.

Choice B is incorrect and may result from subtracting 5, not dividing by 5, on both sides of the equation $5(x-5) = 10$. Choice C is incorrect. This is the value of x , not the value of $x-5$.

Choice D is incorrect. This is the value of $x+5$, not the value of $x-5$.

Q8**1**

The correct answer is 1.2. One way to solve the equation $2(p+1)+8(p-1)=5p$ is to first distribute the terms outside the parentheses to the terms inside the parentheses: $2p+2+8p-8=5p$. Next, combine like terms on the left side of the equal sign: $10p-6=5p$. Subtracting $10p$ from both sides yields $-6=-5p$. Finally, dividing both sides by -5 gives $p=\frac{6}{5}$, which is equivalent to $p=1.2$. Note that 1.2 and $\frac{6}{5}$ are examples of ways to enter a correct answer.

Q9**C**

C. Infinitely many

Choice C is correct. If the two sides of a linear equation are equivalent, then the equation is true for any value. If an equation is true for any value, it has infinitely many solutions. Since the two sides of the given linear equation $66x = 66x$ are equivalent, the given equation has infinitely many solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10

C

C. $950 + 50(t - 3) = 1,150$

Choice C is correct. It's given that the cost to rent a bus is \$ 950 for the first 3 hours and an additional \$ 50 per hour for each hour after the first 3 hours. It's also given that t represents the total number of hours and $t > 3$. Therefore, the number of additional hours after the first 3 hours can be represented by the expression $t - 3$. The cost for these additional hours is \$ 50 per hour, so the cost for the additional hours can be represented by the expression $50(t - 3)$. The total cost can be calculated by adding the cost for the first 3 hours to the cost for the additional hours and can be represented by the expression $950 + 50(t - 3)$. It's also given that the total cost to rent the bus from the company for t hours is \$ 1,150. Thus, the equation that represents this situation is $950 + 50(t - 3) = 1,150$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q5 — the source CB data omitted a structured key for these items, so the answer slot shows —. The worked solution in the rationale below each entry contains the correct value.

Generated 2026-05-29 · cb-educator-question-bank